

Problema 1

```
for (i = n/2; i <= n; i++)  
  for (j = 1; j+n/2 <= n; j++)  
    for (k = 1; k <= n; k = k*2) counter++;
```

Paso 1 ciclo i

$$i = n/2, \dots, n$$

$$\text{vueltas} = n - n/2 + 1 = n/2 + 1 \quad \rightarrow O(n)$$

Paso 2 ciclo j

$$j + n/2 \leq n \Rightarrow j \leq n/2$$

$$j = 1, \dots, n/2$$

$$\text{vueltas} = n/2 \quad \rightarrow O(n)$$

Paso 3 ciclo k

$$k = 1, 2, 4, 8, \dots \Rightarrow k = 2^m$$

$$2^m > n \Rightarrow m > \log_2(n)$$

$$\text{vueltas} = \log_2(n) + 1 \quad \rightarrow O(\log n)$$

Paso 4 estan anidados $\Rightarrow$ se multiplican

$$T(n) = (n/2 + 1) (n/2) (\log_2 n + 1)$$

$$= n^2 \log_2(n) / 4 + n^2/4 + n \log_2(n) / 2 + n/2$$

$$\text{dominante: } n^2 \log_2(n) / 4$$

$$T(n) = O(n^2 \log n)$$

comprobacion P1 (counter)

$$n = 100 \quad 51 \times 50 \times 7 = 17,850$$

$$n = 1,000 \quad 501 \times 500 \times 10 = 2,505,000$$

$$n = 10,000 \quad 5001 \times 5000 \times 14 = 350,070,000$$

Problema 2

if ($n \leq 1$) return n ;

for ($i = 1; i \leq n; i++$)

for ($j = 1; j \leq n; j++$) { printf; break; }

Paso 1 guarda

if ($n \leq 1$) return

-> $O(1)$

Paso 2 ciclo i

$i = 1, \dots, n$

vuelatas = n

-> $O(n)$

Paso 3 ciclo j

$j = 1$ -> printf -> break

vuelatas = 1 (no n)

-> $O(1)$

Paso 4 $T(n) = O(1) + n \times 1 = n + 1$

$T(n) = O(n)$

ojo: son 2 for anidados pero NO es $O(n^2)$

el break corta el ciclo interno

imprime "Sequence" exactamente n veces

Problema 3

```
for (i = 1; i <= n/3; i++)  
    for (j = 1; j <= n; j += 4) printf;
```

Paso 1 ciclo i

$$i = 1, \dots, n/3 \quad \text{vueltas} = n/3 \quad \rightarrow O(n)$$

Paso 2 ciclo j

$$j = 1, 5, 9, \dots \Rightarrow j = 1 + 4m$$

$$1 + 4m > n \Rightarrow m > (n-1)/4$$

$$\text{vueltas} = (n-1)/4 + 1 \sim n/4 \quad \rightarrow O(n)$$

Paso 3 están anidados $\Rightarrow$ se multiplican

$$T(n) = (n/3)(n/4) = n^2 / 12$$

$T(n) = O(n^2)$

comprobación

$n = 1,000$	333×250	$= 83,250$	$n^2/12 = 83,333$
$n = 10,000$	3333×2500	$= 8,332,500$	$n^2/12 = 8,333,333$

Resumen

Problema 1 $\rightarrow O(n^2 \log n)$

Problema 2 $\rightarrow O(n)$

Problema 3 $\rightarrow O(n^2)$